

Theory Appendix to “Automation, Capital Taxation, and Public Insurance”

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How to read the theory appendix

The main paper presents the public-insurance argument in economic order: resource exposure, relative valuation, payoff span, and implementation. This appendix supplies the longer derivations behind that argument. It is organized for a reader who wants to check where ownership, timing, constraints, and approximation enter, rather than for a reader following the chronology in which the model was solved.

The first module closes the Brownian economy from its sequential accounts. The second derives the fiscal stochastic discount factor, the traded-direction valuation condition, and the payoff projection used in Sections 3 and 4. The third develops the deterministic, small-risk, and local-quadratic approximations. The final module states the recursive commitment and solvency conditions that delimit those characterizations. A separate Extended Derivations volume preserves the complete primitive-to-local-system chain in original economic coordinates; it is not required for following the claim-organized proofs here.

The appendix preserves the force of the body statements. A first-order condition is not promoted into existence or sufficiency; an exact projection is kept separate from an order- ϵ^2 welfare comparison; and a local stabilizing system is not used as a global fiscal-feasibility theorem. The common inherited state is (S_0, M_0) . Whenever a transformed or matrix representation is useful, the underlying economic object and component order appear first.

Main-paper object	Appendix task	Principal boundary
Worker, owner, and public accounts	Establish ownership, timing, clearing, and the recursive state	Smooth branch; no global existence claim
Fiscal and market valuation	Derive the two SDFs and the portfolio KKT condition	Necessary condition, with boundary residuals
Payoff span and welfare	Separate exact projection from local welfare value	Public payoff completion is not household-market completion
Tax and capital adjustment	Derive the forward-looking tax identity and local propagation	No universal tax sign or global capacity frontier

T1 Brownian environment, accounts, and conditional recursive interface

This appendix supplies the sequential closure behind Section 2 of the main paper. Its task is deliberately narrower than proving a global Ramsey theorem. It records the technology, ownership, budgets, pricing, and continuation restrictions needed to audit the economic objects in the main text. In particular, a formal HJB written on a smooth interior region is a representation of a candidate recursive problem; it does not establish nonempty admissible sets, a classical solution,

verification, uniqueness, or a global solvency frontier.

T1.1 Technology, ownership, and timing

The filtration is generated by the Brownian pair (B^z, B^x) stated in the main paper. The two shocks have continuous paths and are observed before adapted controls for subsequent instants are chosen. The risky holding used in the stochastic integral must be predictable. Installed capital, the tax rate, and public net worth have distinct timing: K_t and τ_t are continuous; N_t has a Brownian diffusion through the inherited financial position; and the government can rebalance the claim with the purchase recorded in N_t . Hence an innovation does not permit a contemporaneous capital levy or a retroactive choice of its risky payoff.

For completeness, the firm solves the static allocation problem

$$Y(z, \alpha, K, L) = \max_{K_A, K_N, L_A, L_N \geq 0} e^z \left(\frac{e^{\nu_A} K_A + L_A}{\alpha} \right)^\alpha \left(\frac{e^{-\nu_N} K_N + L_N}{1 - \alpha} \right)^{1-\alpha} \quad (\text{T1.1})$$

subject to $K_A + K_N = K$ and $L_A + L_N = L$. The subscripts A and N in this display denote task composites; they are unrelated to public net worth N_t . The three global static regimes are determined by

$$s_A = \log(K/L) - \text{logit}(\alpha) + \nu_A, \quad s_N = \nu_N - \log(K/L) + \text{logit}(\alpha). \quad (\text{T1.2})$$

When $s_A > 0$ and $s_N > 0$, capital is used in the automation-intensive composite and labour in the new-task composite, giving the smooth production formula used in the main paper. If either inequality fails, the appropriate corner formula from the firm problem, rather than the smooth branch formula, determines factor prices. The main text keeps the branch assumption visible because later derivatives use it.

For reference, the global static value is

$$Y = \begin{cases} e^z(e^{\nu_A} K + L), & s_A \leq 0, \\ e^{z+\nu_A\alpha}\Omega(\alpha)K^\alpha L^{1-\alpha}, & s_A \geq 0, s_N \geq 0, \\ e^{z+\nu_A\alpha-\nu_N(1-\alpha)}(K + e^{\nu_N} L), & s_N \leq 0. \end{cases} \quad (\text{T1.3})$$

Adjacent formulas agree at their common boundary. The allocation is homogeneous of degree one and continuously differentiable in (K, L) , although second derivatives can kink at a specialization boundary. The appendix preserves a route for checking whether a proposed path leaves the maintained branch; full-specialization derivatives must not be extrapolated through a change in task allocation.

The global program implies that raising the available automation share while holding (z, K, L) fixed weakly raises optimized output. That fact does not determine the wage response: along

the smooth branch the wage bill is $(1 - \alpha)Y$, so displacement and the induced scale effect work in opposite directions. Nor does it determine the external claim payoff. This is the primitive separation among production, incidence, and financial spanning on which the paper relies.

Private accounts and the domestic capital law. Workers solve no portfolio problem because their asset holdings are identically zero. Their consumption identity is $c_t = W_t + T_t$, with $T_t \geq 0$. This restriction is an institutional feature of the maintained environment: it rules out private hedging by workers, not consumption smoothing by a representative household.

Domestic owners hold installed capital and cannot sell it to the government or to the international investor block. With reversible one-for-one installation and logarithmic preferences, the installed-capital price is $q^D = 1$ and their optimal consumption is $C_t^K = \rho K_t$. Combining their budget with the resource constraint gives

$$dK_t = K_t g(z_t, x_t, K_t, \tau_t) dt, \quad g = (1 - \tau)(R^K - \delta) - \rho. \quad (\text{T1.4})$$

Thus $[K]_t = [K, B^z]_t = [K, B^x]_t = 0$. The capital tax affects workers indirectly through the future rental base, capital stock, and wage; it does not reallocate ownership of the existing domestic capital stock to the public sector. Reintroducing a variable installed-capital price would add private revaluation and change the state system, so it is outside the maintained branch rather than a small robustness variation.

The statutory source tax applies to signed net rental income,

$$\mathcal{B} = (R^K - \delta)K, \quad \tau \leq 1, \quad d\tau = \nu dt, \quad \Psi_\tau = Y\nu^2/(2\kappa_\tau). \quad (\text{T1.5})$$

The base can be negative. At $\tau = 1$ the tax-speed control must point into the admissible set. No conclusion about a globally optimal tax path follows from this local accounting statement.

The tax does not enter the static firm problem directly. Firms choose current factor use from producer rental and wage schedules. The tax changes the owners' after-tax return, which changes the drift of K in (T1.4); future factor prices and the future tax base then change with K . This separates a source tax on rental income from a direct tax on a contemporaneous Brownian claim payoff.

World prices, external trading, and public accounting. The external pricing block is exogenous to domestic policy. For an external claim with return loading $(\sigma_{Rz}, \sigma_{Rx})'$, the world SDF and total return obey the no-arbitrage relation stated in the main paper. In the specialized foreign log-utility closure, the loading coincides with the market-price-of-risk vector and the excess premium is $\|\lambda_I\|^2$. More generally, no-arbitrage requires only the scalar premium $\sigma'_R \lambda_I$. The manuscript uses the latter representation in the body because it makes clear which object is priced and avoids treating the special equality as a universal property of external claims.

For the maintained foreign reference economy, let $\ell_I(z, x)$ be international capitalist log consumption and define

$$\mathcal{L}^{z,x}\phi = \kappa_z(\bar{z} - z)\phi_z + \kappa_x(\bar{x} - x)\phi_x + \frac{\sigma_z^2}{2}\phi_{zz} + \frac{\sigma_x^2}{2}\phi_{xx}. \quad (\text{T1.6})$$

The specialized market-price vector and safe rate are

$$\lambda_I = \begin{pmatrix} \sigma_z \\ A'(x)h_I(\alpha)\sigma_x \end{pmatrix}, \quad r = \rho + \mathcal{L}^{z,x}\ell_I - \frac{1}{2}\|\lambda_I\|^2, \quad (\text{T1.7})$$

where $h_I(\alpha)$ is the maintained international factor-intensity expression. The main text does not need this construction to use a priced external payoff. It makes the specialized equality $\sigma_R = \lambda_I$ recoverable and makes clear that domestic fiscal policy determines neither object.

Foreign-residual absorption closes a nonzero public position without changing the world price process: the domestic country is infinitesimal relative to the external market, or an outside sector takes the offsetting position at the stated SDF prices. This is a small-open closure, not a claim that domestic policy has no real effects. It leaves domestic wages, rental income, capital accumulation, transfers, and public net worth endogenous.

Writing the public safe position as $N - s$, the government account is

$$dN = \left[rN + s\sigma'_R\lambda_I + \tau\mathcal{B} + W - c - \frac{Y}{2\kappa_\tau}\nu^2 \right] dt + s\sigma'_R dB. \quad (\text{T1.8})$$

Substitution of $c = W + T$ yields the equivalent cash-flow expression

$$dN = \left[rN + s\sigma'_R\lambda_I + \tau\mathcal{B} - T - \frac{Y}{2\kappa_\tau}\nu^2 \right] dt + s\sigma'_R dB. \quad (\text{T1.9})$$

The equality is useful as an accounting check. It does not decompose a policy effect into separable causal channels: changes in tax rates, capital, transfers, and positions are jointly determined in a Ramsey allocation.

The maintained portfolio restrictions are

$$-\bar{S}^G \leq s \leq N + \bar{B}^G. \quad (\text{T1.10})$$

The upper bound limits gross safe borrowing because the safe position is $N - s$. Together the bounds require $N \geq -(\bar{B}^G + \bar{S}^G)$ for a nonempty instantaneous portfolio interval. This elementary condition is not the continuation-solvency frontier. A viable continuation must also respect the stochastic law of motion, transfers, tax constraints, no-Ponzi restrictions, and feasible policies from every reached state.

There are three distinct notions here. The accounting stock N may be negative. The instant-

neous nonempty-portfolio condition follows from (T1.10). Continuation solvency asks whether a policy can keep transfers, taxes, positions, and future obligations feasible along the support of the stochastic process. Only the third could support a meaningful debt-capacity statement; neither current net worth nor a later present-value object supplies that frontier by itself.

T1.2 Equilibrium and the Ramsey continuation

Given a policy (c, s, ν) and an inherited state, a competitive equilibrium is defined in the competitive-equilibrium definition stated in the main paper. The following additional conditions are what make the definition suitable for a conditional recursive representation:

1. controls are adapted, the risky position is predictable, and stochastic integrals and discounted utility are integrable;
2. $K > 0$, $c > 0$, $T \geq 0$, the tax and portfolio constraints hold, and the specialization branch is used only while its inequalities hold;
3. private budgets, the public account, goods and factor feasibility, productive ownership, and foreign-residual asset clearing are satisfied;
4. public no-Ponzi, transversality, and continuation-viability conditions apply after every finite stopping time; and
5. the admissible continuation class is stable under restriction and measurable pasting.

The last two clauses carry the commitment convention. Subject to them, the continuation of an admissible date-zero plan can be represented by the current inherited state (S_t, M_t) . In the Brownian benchmark, private blocks generate no promise state, so There is no additional protected promise in the baseline. This is a conditional dynamic- programming interface; it should not be read as a proof that all histories have a unique feasible continuation.

The conditional qualification matters for state sufficiency. The five Brownian states are sufficient only on the maintained continuation class. If a policy protected owner utility, grandfathered a claim, or imposed a history-dependent institutional obligation, the associated promise or multiplier would enter M_t . If equilibrium selection needed a branch label that could not be recovered from the displayed state, that label would augment S_t . The empty- M_t convention simplifies this benchmark; it does not prove that promises are irrelevant in public-finance environments generally.

On a smooth interior promise-neutral region, the continuation value can be written as

$V(z, x, K, N, \tau)$. Its formal recursive equation is

$$\begin{aligned} \rho V = \max_{c, s, \nu} & \left\{ \log c + \mathcal{L}^{z, x} V + KgV_K + \nu V_\tau \right. \\ & + \left[rN + s\sigma'_R \lambda_I + \tau \mathcal{B} + W - c - \frac{Y}{2\kappa_\tau} \nu^2 \right] V_N \\ & \left. + s(\sigma_z \sigma_{Rz} V_{Nz} + \sigma_x \sigma_{Rx} V_{Nx}) + \frac{1}{2} s^2 \|\sigma_R\|^2 V_{NN} \right\}. \end{aligned} \quad (\text{T1.11})$$

The first line propagates exogenous states and installed capital; the second values public cash flow; the third is the covariance block created by the external claim. The valuation section of the main paper uses precisely that block to compare fiscal and market valuation. At transfer, tax, portfolio, specialization, or continuation boundaries, the displayed interior maximization is replaced by the relevant KKT condition or variational inequality. Numerical clipping of an unconstrained policy does not supply the missing boundary economics.

What this module establishes. The appendix makes the main paper's environment closure and the formal recursive object auditable. It does not prove existence of the admissible correspondence, global solvency, a classical HJB solution, verification, unique equilibrium selection, or a global Ramsey optimum. Those are distinct proof and boundary obligations for the claims that use the recursive representation.

Level-state controls and the Ramsey object. The recursive formulation is most transparent when the economic objects are kept in levels. Let $S = (z, x, K, N, \tau)$ and let a policy be a progressively measurable triple $\Pi = (c, s, \nu)$, with s predictable. The controls have different units: c is a flow of worker consumption, s is the market value of the external risky position, and ν is a rate of change of the statutory tax rate. The policy is admissible only if it can be paired with a competitive equilibrium in the sense of the competitive-equilibrium conditions; a control path by itself does not close the economy.

For a given inherited augmented state (S_0, M_0) , write $\mathcal{CE}(\Pi; S_0, M_0)$ for the associated equilibrium correspondence. The set of admissible policy–equilibrium pairs is

$$\begin{aligned} \mathfrak{R}(S_0, M_0) = \{ (\Pi, E) : E \in \mathcal{CE}(\Pi; S_0, M_0), \\ (\Pi, E) \text{ honors } M_0 \text{ and satisfies the continuation restrictions} \}. \end{aligned} \quad (\text{T1.12})$$

The Ramsey value used in the next section is

$$V^R(S_0, M_0) = \sup_{(\Pi, E) \in \mathfrak{R}(S_0, M_0)} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} \log c_t dt \right]. \quad (\text{T1.13})$$

This formulation makes the welfare population explicit: only workers' consumption enters the objective. Domestic owners and international investors remain in the equilibrium constraints, but they do not receive an implicit social weight.

The domain of (T1.13) cannot be described by a rectangular numerical box. At a minimum it requires $K > 0$, $c > 0$, $T = c - W \geq 0$, $\tau \leq 1$, and the portfolio interval in (T1.10). The production formula used in a local calculation also requires the relevant specialization inequalities. In addition, a state must admit a continuation that obeys no-Ponzi, integrability, and transversality conditions. The last requirement is support based: it concerns the set of states that can be reached under the diffusion, not merely the path produced by a single numerical simulation.

These distinctions matter at boundaries. At the transfer floor, worker consumption is constrained by $c \geq W$. At the tax ceiling, the tangent cone restricts the sign of ν . At a portfolio bound, the unconstrained first-order condition is replaced by a KKT inequality. At a specialization boundary, factor-price derivatives must be computed from the global firm program. At a continuation-solvency boundary, several instruments can become jointly restricted. Treating any of these objects as an ordinary terminal value on an artificial box face changes the economic problem.

Sequential accounting in full. The private and public accounts can be read as one sequence. Firms rent capital and labour and pay the producer rental R^K and wage w . The domestic owner receives the after-tax rental flow $(1 - \tau)(R^K - \delta)K$, consumes $C^K = \rho K$, and invests the remainder in installed capital. Workers receive $W = wL$ and the transfer T and consume $c = W + T$. The government receives the source-tax flow $\tau\mathcal{B}$, where $\mathcal{B} = (R^K - \delta)K$, pays transfers and tax-adjustment costs, and finances any shortfall with its safe and risky financial positions. International investors price the external securities and absorb the residual position. No line in this sequence turns the private wage bill into a Treasury asset.

Writing the external return as

$$dR = (r + \beta_R)dt + \sigma_{Rz} dB^z + \sigma_{Rx} dB^x, \quad \beta_R = \sigma_{Rz}\lambda_{Iz} + \sigma_{Rx}\lambda_{Ix}, \quad (\text{T1.14})$$

makes the component order explicit before the vector notation is used. A government with risky position s and safe position $N - s$ therefore has

$$\begin{aligned} dN &= (N - s)r dt + s dR + \tau\mathcal{B} dt - T dt - \frac{Y}{2\kappa_\tau}\nu^2 dt \\ &= \left[rN + s\beta_R + \tau\mathcal{B} - T - \frac{Y}{2\kappa_\tau}\nu^2 \right] dt + s\sigma_{Rz} dB^z + s\sigma_{Rx} dB^x. \end{aligned} \quad (\text{T1.15})$$

Equation (T1.15) is equivalent to the main-text budget. Its component form adds one useful fact: the government can scale the two shock exposures together with s , but cannot select their ratio. That fixed ratio is the source of the rank-one span restriction later in the paper.

The budget also clarifies the usual short-window accounting. Conditional on the state at the beginning of a window of length Δ , bounded tax and transfer flows are of order Δ . The inherited risky position produces the random term

$$s_{t-} \left(\sigma_{Rz} \sqrt{\Delta} \varepsilon_z + \sigma_{Rx} \sqrt{\Delta} \varepsilon_x \right) + o_p(\sqrt{\Delta}). \quad (\text{T1.16})$$

The display is a timing identity, not a claim that the government cannot reset policy after observing an innovation. It says only that an adjustment selected after the innovation cannot be the holding that generated its realized return. It also does not turn a physical expected excess return into market-valued fiscal income: under the world SDF, the excess return compensates risk exposure.

Task allocation and inherited-capital incidence. The compact production representation in the main text is useful precisely because the full allocation has already been specified here. The firm assigns capital and labour to two composites with different factor intensities. When capital is scarce, labour is also used in the automation-intensive composite. When capital is abundant, capital also enters the new-task composite. Between those cases, full specialization holds and the task share α maps transparently into output and factor income. The two margins in (T1.2) distinguish full use of available automation from full factor specialization; neither should be replaced by the phrase “high automation.”

At inherited K and L , the smooth branch delivers

$$\frac{\partial \log Y}{\partial x} = A'(x) s_A, \quad \frac{\partial \log W}{\partial x} = A'(x) \left(s_A - \frac{1}{1-\alpha} \right). \quad (\text{T1.17})$$

The scale effect is positive on this branch, while the direct fall in labour’s share opposes it in the wage equation. Thus a negative impact wage effect is compatible with a positive output effect whenever $0 < s_A < 1/(1-\alpha)$. This is a local incidence condition under the stated technology and inherited capital; it is not a forecast, a causal estimate, or a statement about the long-run wage response.

Capital accumulation supplies the main opposing dynamic force. At a fixed tax rate, the higher rental return can increase future K , which changes later output and wages. The source model’s deterministic steady-state comparison can therefore produce a positive long-run wage derivative even when the inherited-capital derivative in (T1.17) is negative. The paper uses this difference to organize impact and propagation, not to claim that wage recovery is universal. Constraint activation, a different installation technology, or a path that leaves full specialization changes the relevant calculation.

The claim-neutral mechanism case can now be written more precisely. Let a local primitive direction (dz, dx) satisfy

$$\sigma_{Rz} dB^z + \sigma_{Rx} dB^x = 0 \quad \text{while} \quad dW \neq 0 \text{ or } d\mathcal{B} \neq 0. \quad (\text{T1.18})$$

Along this direction the external return has no instantaneous loading, even though worker resources or the fiscal base move. The condition is a negative control for the asset, not a claim that tax policy is powerless: taxes can still alter later flows and capital accumulation. Its purpose is to rule out the false inference that an asset with an automation label necessarily spans automation-relevant fiscal exposure.

The formal recursion term by term. For a smooth value function $V(z, x, K, N, \tau)$, define the exogenous-state generator

$$\mathcal{L}^{z,x}V = \kappa_z(\bar{z} - z)V_z + \kappa_x(\bar{x} - x)V_x + \frac{\sigma_z^2}{2}V_{zz} + \frac{\sigma_x^2}{2}V_{xx}. \quad (\text{T1.19})$$

The formal interior HJB in (T1.11) has five economic blocks. The flow utility $\log c$ records worker welfare. The generator in (T1.19) moves the two exogenous technologies. The term KgV_K moves inherited productive capital through private saving. The term νV_τ records gradual tax adjustment. The coefficient on V_N values the drift of public financial resources. Finally, the two remaining terms are generated because N and (z, x) share Brownian innovations through the external claim.

To see the last block without matrix notation, define

$$D[V] = \sigma_z \sigma_{Rz} V_{Nz} + \sigma_x \sigma_{Rx} V_{Nx}, \quad q_R = \sigma_{Rz}^2 + \sigma_{Rx}^2. \quad (\text{T1.20})$$

Then $sD[V]$ is the cross-variation between financial wealth and the two technology states, and $s^2 q_R V_{NN}/2$ is the curvature cost or benefit of the induced financial risk. The portfolio condition in the valuation section of the main paper is generated by this block, not by the wage equation or a generic preference for equity. The appendix states the components so that the subsequent vector projection has a recoverable economic origin.

At a smooth interior point with $V_N > 0$, $V_{NN} < 0$, and $q_R > 0$, the consumption and tax-speed first-order conditions are

$$c = \frac{1}{V_N}, \quad \nu = \frac{\kappa_\tau}{Y} \frac{V_\tau}{V_N}. \quad (\text{T1.21})$$

They are local characterizations. The first is superseded by the transfer floor if it would imply $c < W$; the second is superseded at the tax ceiling and any other active constraint. The portfolio first-order condition is deferred to the valuation appendix because its reader-facing role is to compare fiscal and market valuation, not to close the present environment. Keeping that derivative out of the environment section is a matter of economic order, not an omission of the public portfolio from the model.

T1.3 Normalization and reduced coordinates

Some later local calculations divide by installed capital. Define, only for that purpose,

$$k = \log(K/L), \quad n = N/K, \quad \tilde{c} = c/K, \quad \theta = s/K. \quad (\text{T1.22})$$

These are transformations, not new economic primitives: k is log capital per worker, n is public net worth per unit of installed capital, $\tilde{c}$ is worker consumption per unit of capital, and θ is the risky dollar position per unit of capital. Since K has finite variation, no Itô correction is created by this normalization. The normalized public account is

$$dn = \left[(r - g)n + \theta\beta_R + f - \tilde{c} - \frac{y}{2\kappa_\tau}\nu^2 \right] dt + \theta\sigma_{Rz} dB^z + \theta\sigma_{Rx} dB^x, \quad (\text{T1.23})$$

where $y = Y/K$, $f = (\tau\mathcal{B} + W)/K$, and $g = \dot{K}/K$. These coordinates are useful for local approximation and computation because they separate scale from composition. They do not make n stationary, replace the level-state feasibility problem, or turn θ into a portfolio share with a fixed denominator.

Commitment tails. The maintained continuation convention is date-zero commitment with a state-based admissible tail. Suppose the correspondence in (T1.12) is nonempty on a region, stable under restriction and measurable pasting, and its relevant equilibrium selection is recoverable from (S, M) . Under those conditions, restricting an admissible date-zero plan after a finite stopping time yields a feasible continuation from the inherited state at that time, and a feasible continuation can be pasted onto the original prefix. The recursive value then represents the value of the unchanged commitment tail.

The qualification is substantive. It does not establish a globally viable domain, measurable selection, a classical solution, or uniqueness of the recovered continuation controls. A numerical solution on a compact box also cannot establish these objects. For the paper's conditional analytical results, the necessary statement is narrower: the displayed state, budgets, and admissibility conventions identify the environment in which an interior fiscal-valuation calculation is being interpreted. The formal proof and any claim of global implementation require the separate assurance and boundary work.

The five-state equation. For local analysis, divide variables by installed capital after the level economy has been defined. Let $k = \log(K/L)$, $n = N/K$, $\tilde{c} = c/K$, and $\theta = s/K$, as in (T1.22). The associated output, wage, rental-base, fiscal-flow, and capital-growth maps are

$$y = e^{z+\nu_A\alpha}\Omega(\alpha)e^{(\alpha-1)k}, \quad w = (1-\alpha)y, \quad b = \alpha y - \delta, \quad f = \tau b + w, \quad g = (1-\tau)b - \rho. \quad (\text{T1.24})$$

The notation preserves units: y , w , and b are flows per unit of installed capital; n and θ are balance-sheet quantities per unit of capital, not shares with a common fixed denominator. Because K has finite variation, the normalized state system has no extra quadratic-variation correction.

The normalized wealth equation is

$$dn = \left[(r - g)n + \theta\beta_R + f - \tilde{c} - \frac{y}{2\kappa_\tau}\nu^2 \right] dt + \theta\sigma_{Rz} dB^z + \theta\sigma_{Rx} dB^x. \quad (\text{T1.25})$$

It makes the three instruments readable in the same units. Safe finance appears in $(r - g)n$, risky finance in the drift premium and two diffusion terms multiplied by θ , and taxation in f , g , and the adjustment cost. The reduction does not remove the tax state or the continuation-viability problem: (z, x, k, n, τ) remains a five-dimensional state.

Let the normalized value be $U = (\log L + k)/\rho + v(z, x, k, n, \tau)$. For the smooth interior branch, define

$$D_v = \sigma_z\sigma_{Rz}v_{zn} + \sigma_x\sigma_{Rx}v_{xn}, \quad q_R = \sigma_{Rz}^2 + \sigma_{Rx}^2. \quad (\text{T1.26})$$

The resulting formal equation is

$$\begin{aligned} \rho v = & -\log v_n - 1 + \mathcal{L}^{z,x}v + g\left(\frac{1}{\rho} + v_k\right) + [(r - g)n + f]v_n \\ & + \frac{\kappa_\tau v_\tau^2}{2y v_n} - \frac{(\beta_R v_n + D_v)^2}{2q_R v_{nn}}. \end{aligned} \quad (\text{T1.27})$$

This representation is equivalent to the level HJB on its stated domain. It is not a claim that the economic problem has a stationary wealth share or that the public-wealth direction is mean reverting. In the specialized foreign reference closure, $q_R = \beta_R$, but generic pricing uses $\beta_R = \sigma'_R \lambda_I$ and keeps the variance q_R distinct.

The interior feedback rules are

$$\tilde{c} = \frac{1}{v_n}, \quad \nu = \frac{\kappa_\tau v_\tau}{y v_n}, \quad \theta = -\frac{\beta_R v_n + D_v}{q_R v_{nn}}. \quad (\text{T1.28})$$

They require $v_n > 0$, $v_{nn} < 0$, and the slack transfer, tax, portfolio, specialization, and continuation constraints already listed. The third expression is included to show where later valuation work enters: it contains an ordinary priced return term and a cross-variation term. It is not interpreted here as a globally feasible portfolio policy.

Why fiscal wealth does not collapse the state. In a deterministic frozen-state benchmark, a present value of future fiscal flows can be added to public financial net worth. The analogous stochastic shortcut generally fails. Consider a shifted-wealth candidate

$$v = \frac{1}{\rho} \log[n + j(z, x, k, \tau)] + \omega(z, x, k, \tau). \quad (\text{T1.29})$$

The diffusion exposure of j is

$$\zeta_j = (\sigma_z j_z, \sigma_x j_x)'. \quad (\text{T1.30})$$

One external claim can offset only its component parallel to σ_R . The remaining component,

$$\zeta_j^\perp = \zeta_j - \sigma_R \frac{\sigma_R' \zeta_j}{\|\sigma_R\|^2}, \quad (\text{T1.31})$$

leaves a curvature term proportional to $\|\zeta_j^\perp\|^2/(n+j)^2$ after the marketed direction is optimized. That term cannot generally be folded into an equation for j alone. The economic reason is the same as in the main text: domestic fiscal income is additive to public wealth and has risk outside the available payoff span. The failure of scalar separation is not a numerical inconvenience; it is the mathematical form of the paper's incomplete-span mechanism.

The deterministic fiscal benchmark. When common states are frozen and volatility is removed, define

$$J(K, \tau) = \sup_{\nu} \int_0^\infty e^{-\rho t} \left[\tau \mathcal{B} - \frac{Y}{2\kappa_\tau} \nu^2 \right] dt, \quad (\text{T1.32})$$

subject to the deterministic laws for K and τ . The object J is deferred to the fiscal-wealth section of the paper because its role is to explain a specific deterministic and small-risk comparison. It is not a debt limit, it is not a claim on workers' private wages, and it does not make the stochastic recursion one-dimensional. Once Brownian risk is restored, the residual in (T1.31) and any active constraints obstruct the deterministic decomposition.

T1.4 Economic boundaries and recoverability

The main paper keeps the economic consequences of boundaries near the corresponding claim. This appendix records their formal location. The production boundary replaces the smooth factor-price map by the global program in (T1.1). The transfer boundary changes the consumption first-order condition. The tax boundary changes the admissible tangent set for ν . The portfolio bounds change the risky-position derivative into a KKT condition. The continuation-solvency boundary requires diffusion tangency and inward drift for the actual constrained state process. These are separate objects; a broad numerical guard rail or a positive value of N cannot serve as a generic substitute for all of them.

For a smooth normalized solvency frontier written schematically as $n = \underline{n}(z, x, k, \tau)$, diffusion tangency must be checked against both Brownian directions, and the remaining drift must point inward under feasible controls. A local reference floor can demonstrate viability on a declared compact region, but it does not characterize the global frontier. Accordingly, the environment supports a conditional interior interpretation of later equations and makes boundary obligations auditable; it does not turn a computed interior path into a global implementation result.

Clearing in the small open economy. The small-open closure is sometimes misread as eliminating equilibrium restrictions. It eliminates domestic feedback to the world SDF and the world supply curve for the external claim. It does not eliminate domestic feasibility. At every date, firms' factor demands exhaust the installed domestic capital stock and fixed labour supply. Domestic owners retain the productive claim, workers receive their wage bill, and the government's transfer is financed by its tax and financial accounts. The external risky position held by the government has an offsetting foreign position; that asset clearing condition is distinct from ownership of domestic capital.

The distinction is especially important for incidence. The gain in a domestic owner's private capital income is not automatically a Treasury cash flow, and an increase in the market value of an external claim is not automatically a gain in domestic output. The public account records the latter only for the position actually held by the government. Conversely, the worker's private wage is not a public asset merely because the government values worker consumption. The public-finance problem arises from this separation of private ownership, fiscal resources, and traded payoffs.

The same distinction governs the resource cost of tax adjustment. The term $Y\nu^2/(2\kappa_\tau)$ is paid in final goods. It reduces the resources available to the domestic economy; it is not a financial transaction between the government and capital owners. The source-tax transfer changes the division of rental income, while the adjustment cost makes rapid changes in that division costly. Thus the government cannot make an arbitrary state-contingent transfer by sending ν to infinity, even before portfolio and continuation constraints are considered.

Information and control recovery. All policies are adapted to the Brownian filtration, but the risky position in the stochastic integral is predictable. This is the continuous-time expression of a simple economic timing rule: the portfolio that earns the payoff over a short shock window was selected before that window's innovation is observed. The policy can be rebalanced immediately afterwards, with the trade booked at the prevailing market price, but the new position earns only future return increments. Capital and the tax rate carry still stronger inheritance because their laws have finite variation.

The distinction matters for recovery from a recursive policy. A candidate value function must yield a feasible c , s , and ν at each state, and the resulting objects must be placed back into the worker budget, owner accumulation law, public account, firm allocation, and external market-clearing convention. Solving a formal maximization over c , s , and ν without this recovery step does not establish a Ramsey allocation. In particular, a portfolio coefficient must be converted to a dollar holding, checked against its bounds, and used in the same SDF process that priced the comparison. A tax-speed coefficient must be integrated into a rate that respects the ceiling and the inherited initial rate.

The maintained promise-neutral convention also has a timing interpretation. It says that the date-zero state carries no protected private continuation value or contract whose promised value

must be delivered after a shock. If such a commitment were added, the policy could not simply be recovered from (z, x, K, N, τ) : its promised value, multiplier, or branch label would have to be stored in M . This is why the notation The augmented-state notation is retained for applications with protected promises; the baseline has no such additional object.

The interior HJB in economic units. The formal HJB is useful only if every term maps back to the sequential system. The generator $\mathcal{L}^{z,x}$ is generated by the stated OU laws. The term KgV_K is generated by owners' saving, not government investment. The term νV_τ is generated by the tax-rate law. The coefficient on V_N is exactly the drift in the public-net-worth SDE. The cross term $sD[V]$ arises because a public claim payoff and the technology states share Brownian innovations. The quadratic term in s arises because public net worth itself becomes risky when the government holds the claim.

This map supplies several quick sanity checks. If $s = 0$, the public account has no financial diffusion and the covariance and quadratic terms vanish. If the claim has only one nonzero component, its payoff can load on only that Brownian direction. If $\nu = 0$, the tax rate is frozen but the tax flow can still change with rentals and capital. If K is inherited, an innovation can change g but cannot create a diffusion term in K . These limits are useful checks on prose and algebra alike; none is a proof that the resulting constrained policy is globally optimal.

What this module supplies. This module gives the environment and valuation sections of the main paper a common economic closure. The next theory module may take as defined the worker welfare object, domestic ownership structure, external price system, public-net-worth law, and conditional recursive state. It must still derive the fiscal SDF and the portfolio valuation condition from the appropriate marginal value. The theory here does not license a claim of complete markets, complete insurance, a natural debt limit, or a global solution.

T2 Fiscal Valuation, Payoff Projection, and the Small-Risk Expansion

Reader map. The valuation and payoff-span sections of the main paper use one conditional Brownian mechanism: a public claim can change fiscal exposure only along its payoff. This appendix verifies the three mathematical links required for that statement: the fiscal marginal value produces a fiscal risk price; the portfolio derivative is the projection of fiscal–market valuation disagreement; and the small-risk position separates return demand from the fiscal hedge. The appendix does not prove global existence, uniqueness, admissibility on an unbounded state space, or a structural debt limit.

T2.1 Fiscal and market prices of risk

Fix the worker-welfare Brownian continuation with inherited state

$$S_0 = (z_0, x_0, K_0, N_0, \tau_0).$$

The derivation assumes a smooth candidate value V , $V_N > 0 > V_{NN}$, a nonzero external-claim loading σ_R , and the small-open pricing relation $\beta_R = \sigma'_R \lambda_I$. It is conditional on the maintained smooth full-specialization branch and on slack transfer, tax, portfolio, solvency, and production-switch constraints. These restrictions matter: a binding constraint replaces an equality by a KKT condition, and a new protection promise would change any additional protected promise inherited at date zero.

From public-net-worth diffusion to a fiscal price of risk. The diffusion of public financial net worth created by the public position is

$$dN_t|_{dW} = s_t \sigma_{Rz,t} dW_t^z + s_t \sigma_{Rx,t} dW_t^x. \quad (\text{T2.1})$$

Capital and the tax rate have finite variation at a Brownian innovation. Itô's formula applied to V_N therefore yields

$$dV_N|_{dW} = [\sigma_z V_{Nz} + s V_{NN} \sigma_{Rz}] dW_t^z + [\sigma_x V_{Nx} + s V_{NN} \sigma_{Rx}] dW_t^x. \quad (\text{T2.2})$$

Discounting by $e^{-\rho t}$ changes the drift but not these coefficients. Comparing with the definition of m_G gives the component fiscal-risk-price identity used in the main paper. The result is a local identity for a candidate continuation. It does not imply that m_G prices every unavailable state claim.

Traded-direction valuation. The portfolio block is the quadratic Hamiltonian displayed in the main paper. Its derivative is

$$\mathcal{H}_s = \beta_R V_N + D[V] + s \|\sigma_R\|^2 V_{NN}. \quad (\text{T2.3})$$

The market-pricing relation replaces the first term by $V_N(\sigma_{Rz} \lambda_{Iz} + \sigma_{Rx} \lambda_{Ix})$. Multiplying the two lines of the component fiscal-risk-price identities by σ_{Rz} and σ_{Rx} respectively and adding replaces the last two terms by

$$D[V] + s \|\sigma_R\|^2 V_{NN} = -V_N(\sigma_{Rz} \lambda_{Gz} + \sigma_{Rx} \lambda_{Gx}). \quad (\text{T2.4})$$

Substitution gives the scalar valuation cancellation displayed in the main paper. This is the load-bearing cancellation: the market premium and the fiscal covariance are not two unrelated motives

once both are expressed in prices of risk.

For an unconstrained candidate, the first-order condition is $\mathcal{H}_s = 0$. For a signed-holding convention with $s \in [\underline{s}, \bar{s}]$, strict local concavity follows from $V_{NN} < 0$ and $\|\sigma_R\|^2 > 0$, giving the one-sided KKT conditions stated in the main paper. If a computation instead parameterizes debt as positive, the inequalities reverse. No numerical output can be read without checking this sign convention and the active set.

T2.2 Projection and local welfare

For the aligned small-risk payoff $\hat{\sigma}_R$, choose h_J as in the main paper's component projection formula. Substitution into its residual definition implies orthogonality. Uniqueness follows because an alternative coefficient satisfying orthogonality would differ by a scalar multiple of $\|\hat{\sigma}_R\|^2$, which is nonzero. Thus a portfolio change can remove the marketed component only by taking its opposite exposure, $-h_J \hat{\sigma}_R$.

In the primitive aligned-payoff normalization,

$$\hat{\sigma}_R = (\hat{\sigma}_z, a(x)h_I(\alpha)\hat{\sigma}_x)', \quad \zeta_J = (\hat{\sigma}_z J_z, \hat{\sigma}_x J_x)'. \quad (\text{T2.5})$$

Full local span is equivalent to

$$J_x = a(x)h_I(\alpha)J_z. \quad (\text{T2.6})$$

This is a relation between the exposure of worker-fiscal resources and the payoff loading. It is not a claim about technology alone, household market access, or implementability.

The small-risk portfolio. On the deterministic separated branch, $X = N + J$ and

$$V_N^0 = \frac{1}{\rho X}, \quad V_{NN}^0 = -\frac{1}{\rho X^2}. \quad (\text{T2.7})$$

Scale both Brownian loadings and prices of risk by ε . Substituting (T2.7) into the order- ε^2 portfolio FOC gives

$$s^0 = -\frac{\hat{\beta}_R V_N^0 + \hat{D}[V^0]}{\|\hat{\sigma}_R\|^2 V_{NN}^0} = X - h_J, \quad (\text{T2.8})$$

where the second equality uses $\hat{\sigma}_R = \hat{\lambda}_I$. The first term is compensated-return demand and the second is the marketed fiscal hedge. Neither term identifies an admissible gross portfolio or a Treasury borrowing limit.

Deterministic fiscal wealth. The small-risk statement begins from a deterministic object rather than from a claim-price approximation. Freeze Brownian risk on the named deterministic characteristic. Let $F(K, \tau)$ denote worker wages plus net rental-tax resources, and let $g(K, \tau)$

be capital growth. Fiscal wealth is the value of choosing the future tax-speed path,

$$J(K, \tau) = \sup_{\{\nu_u\}} \int_0^\infty e^{-\rho u} \left[F(K_u, \tau_u) - \frac{Y(K_u)}{2\kappa_\tau} \nu_u^2 \right] du, \quad \dot{K} = Kg(K, \tau), \quad \dot{\tau} = \nu. \quad (\text{T2.9})$$

The associated interior HJB is

$$\rho J = F + KgJ_K + \frac{\kappa_\tau}{2Y} J_\tau^2. \quad (\text{T2.10})$$

The tax-speed first-order condition is $\nu = \kappa_\tau J_\tau / Y$. It equates the current resource cost of moving the tax rate with the effect of that move on future worker-fiscal resources. It does not say that a high current tax base calls for a high tax rate, because J_K and the rest of the future characteristic also move.

Set $X = N + J$. Subtracting (T2.10) from the deterministic public budget on this interior branch gives

$$\dot{X} = \rho X - c. \quad (\text{T2.11})$$

The log solution in (T2.7) then follows from the standard one-dimensional consumption problem. The derivation uses the common safe-rate closure on the frozen-state benchmark. Deterministic mean reversion adds an opportunity-value term to the level of value; it does not change the zeroth-order consumption rule or the leading portfolio formula. Neither J nor $X > 0$ is a pledgeability statement. The structural fiscal boundary is a separate viability object.

Several claims and the residual. Let $\Sigma = [\hat{\sigma}_{R1}, \dots, \hat{\sigma}_{Rq}]$ have full column rank. The least-squares coefficient that minimizes $\|\zeta_J - \Sigma h\|^2$ solves the normal equations

$$\Sigma' \Sigma h = \Sigma' \zeta_J, \quad h = (\Sigma' \Sigma)^{-1} \Sigma' \zeta_J. \quad (\text{T2.12})$$

The residual $\zeta_J^\perp = \zeta_J - \Sigma h$ satisfies $\Sigma' \zeta_J^\perp = 0$. Expanding the squared norm gives

$$\|\zeta_J\|^2 = \|\Sigma h\|^2 + \|\zeta_J^\perp\|^2, \quad (\text{T2.13})$$

because the cross product vanishes. This is the nonroutine geometric step behind the local welfare comparison. If $\Sigma' \Sigma$ is singular, the displayed inverse is unavailable; the attainable set is still the column space, but a pseudoinverse or a reduced independent menu must be used before assigning positions.

Exact projection and local welfare. The fiscal-SDF identity and the KKT relation are exact conditional identities for the named smooth Brownian continuation. The projection is exact linear geometry for the declared local exposure. The following additional steps are local: scaling by ε , using deterministic J , retaining only the regular order-two welfare term, and treating the deterministic active set as unchanged. A finite-risk calculation could leave that active set, approach

a solvency boundary, or change the continuation policy enough that the expansion is no longer informative. These are logical limits of the result, not numerical niceties.

Failure cases. Several simple cases identify the scope of the formulas. If $\sigma_R = 0$, the claim has no payoff direction and neither the projection coefficient nor the portfolio curvature is identified. With one Brownian shock and one nonzero claim, the local residual vanishes mechanically, but this does not prove fiscal feasibility or household market completion. If V_{NN} fails to be negative, the local portfolio problem need not be concave. If a transfer, tax, portfolio, solvency, or production-switch constraint binds, the relevant equation is a KKT or variational-inequality condition; the separated formula must not be reused without that treatment. Finally, if a different ownership or promise convention adds a protected continuation value, the state and fiscal marginal value have changed. The formulas then require a new derivation rather than a relabelling.

T2.3 Relative valuation and capital taxation

For completeness, apply Itô's quotient rule to $\Theta = m_G/m_I$. The drift contains the safe-rate and quadratic-variation terms and is not used in the instantaneous portfolio condition. The diffusion is

$$\left. \frac{d\Theta_t}{\Theta_t} \right|_{dW} = [(\lambda_{Iz} - \lambda_{Gz})dW_t^z + (\lambda_{Ix} - \lambda_{Gx})dW_t^x]. \quad (\text{T2.14})$$

The return innovation is $dR|_{dW} = \sigma_{Rz}dW^z + \sigma_{Rx}dW^x$. Their instantaneous covariance is consequently

$$\frac{\text{Cov}_t(dR, d\Theta/\Theta)}{dt} = \sigma'_R(\lambda_I - \lambda_G). \quad (\text{T2.15})$$

Equation (T2.15) is the economic reading of the Hamiltonian derivative divided by V_N . A positive covariance calls for a marginal increase in a signed asset holding before constraints are imposed. The statement is about the return payoff, not the name of the asset or a regression of worker wages on asset returns.

Under the maintained log-consumption interpretation, $c = 1/V_N$ implies

$$d \log c|_{dW} = \lambda_{Gz}dW^z + \lambda_{Gx}dW^x, \quad d \log C^I|_{dW} = \lambda_{Iz}dW^z + \lambda_{Ix}dW^x. \quad (\text{T2.16})$$

Thus the interior FOC is orthogonality of the traded return and the innovation in relative worker consumption. It is not a statement that workers and investors have identical SDFs, because only the projection on the available claim is set to zero.

The deterministic tax condition. The fiscal-wealth equation can also be written along a selected deterministic characteristic. Let $D_{t,u}$ denote the discount factor appropriate to that characteristic. Integrating the J_τ costate forward, subject to its terminal condition, gives the conditional

identity

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty D_{t,u} B_u [1 - J_K(S_u)] du. \quad (\text{T2.17})$$

The current tax base enters the integrand, but the object that decides the tax-speed trade-off is the whole future path of the base multiplied by one minus the fiscal value of preserving capital. The identity is conditional on the selected smooth feasible characteristic, positive denominators, the displayed terminal condition, and slack constraints. It is neither an impact tax rule nor a general sign theorem.

At a convergent deterministic interior fixed point with common discounting and a positive net-rental base, the private and fiscal return conditions can be written

$$(1 - \tau)(R^K - \delta) = \rho, \quad \tau(R^K - \delta) = \rho. \quad (\text{T2.18})$$

Adding them produces the conditional one-half benchmark. It is included here only to locate the fiscal-wealth construction: it does not establish convergence, select a public-net-worth level, apply to the AK/Poisson closure, or supply a tax recommendation.

The order-two welfare comparison. Write the regular value expansion as

$$V^\varepsilon = V^0 + \varepsilon^2 V^2 + O(\varepsilon^4). \quad (\text{T2.19})$$

There is no order- ε value term around the mean-zero deterministic reference allocation. The risky position can nevertheless be order one because its expected-return and variance terms both scale with ε^2 . After the portfolio quadratic is completed, the source for V^2 includes the safe-rate correction, curvature of fiscal wealth, the market-return reward, the marketed fiscal covariance, and the residual unspanned exposure. The completion increment derived just below in (T2.21) isolates the last object only under the common-path and unchanged-active-set conditions already stated.

For this reason an exact projection statement and a local welfare statement have different logical status. The residual is exactly outside the current payoff span. The discounted squared-residual expression is a leading welfare object only. A finite-risk result would have to check the continuation policy, state constraints, remainder, and feasibility of the portfolio required by the completion. None of those requirements is supplied by a low residual at one diagnostic node.

Completing the local quadratic yields the one-claim and completion expressions

$$V^{\text{one}} - V^{\text{none}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\|\widehat{\sigma}_{R,t}\|^2}{2\rho} \left(\frac{s_t^0}{X_t} \right)^2 dt + O(\varepsilon^4), \quad (\text{T2.20})$$

$$V^{\text{complete}} - V^{\text{one}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\|\zeta_{J,t}^\perp\|^2}{2\rho X_t^2} dt + O(\varepsilon^4). \quad (\text{T2.21})$$

The formulas require a common feasible deterministic characteristic, a regular expansion, optimizer selection, transversality, and a positive distance from all relevant constraints. They are neither finite-risk numerical welfare results nor certificates of global policy optimality.

T2.4 Scope and verification

The exact-in-deterministic-state R3 calculation reports pointwise order-two corrections only. At the deterministic anchor it records $\nu_2 = -4.71141 \times 10^{-4}$ and $s_2 = -7.72374 \times 10^{-4}$, with reported diagnostic envelopes of 2.66 and 3.50 percent of their absolute effects. These quantities are evaluated on deterministic paths under illustrative inputs. They are not stochastic expected paths, integrated tax responses, finite-risk solutions, calibrated policy effects, or evidence of a general sign. A cheaper R4 truncation reverses the anchor sign of ν_2 and changes the magnitude of s_2 materially; this is why the coefficients remain method diagnostics pending independent reconstruction, common-accounting replay, and claim-specific error evidence.

Mathematical obligations beyond the local result. The traded-direction condition requires a value function differentiable in the states and public net worth, a nondegenerate return loading, and an interior or correctly specified KKT optimum. The leading decomposition additionally requires a regular small-risk expansion, a common feasible deterministic characteristic, and optimizer selection with a positive distance from changes in the active set. Extending either statement to a global theorem would require existence of an admissible continuation, verification for the constrained value problem, and transversality on the relevant unbounded or stopped domain. Giving any numerical coefficient an economic interpretation further requires a common-state comparison, independent control recovery, boundary and refinement checks, and an error envelope narrower than the claimed sign or magnitude.

The generic leading position and the R19 specialization. The specialised formula in the main paper has a transparent generic antecedent. Set $\sigma_R = \varepsilon \hat{\sigma}_R$, $\lambda_I = \varepsilon \hat{\lambda}_I$, and use $V_N^0 = (\rho X)^{-1}$ and $V_{NN}^0 = -(\rho X^2)^{-1}$. The cross derivative is

$$\hat{D}[V^0] = -\frac{\zeta_{Jz} \hat{\sigma}_{Rz} + \zeta_{Jx} \hat{\sigma}_{Rx}}{\rho X^2}. \quad (\text{T2.22})$$

The leading first-order condition therefore gives

$$s^0 = \frac{\hat{\sigma}'_R \hat{\lambda}_I}{\|\hat{\sigma}_R\|^2} X - \frac{\zeta'_J \hat{\sigma}_R}{\|\hat{\sigma}_R\|^2}. \quad (\text{T2.23})$$

The first term is compensated return demand and the second is the marketed fiscal hedge. Under the aligned R19 closure, $\hat{\sigma}_R = \hat{\lambda}_I$, the first coefficient is one and the second is h_J , yielding $s^0 = X - h_J$. This derivation is why the specialised result cannot be silently imported into a different

market-price closure.

Proof and boundary obligations. The exact valuation identity requires a differentiable candidate value, the stated public-net-worth diffusion, predictable trading, and the market-pricing relation. The KKT statement additionally requires a defined feasible interval and local concavity. The deterministic separation requires integrability and its terminal condition. The order-two comparison requires a regular expansion, a common feasible deterministic baseline, unchanged active constraints, and feasibility of each comparison position. The current records do not prove global existence, uniqueness, or the outer solvency frontier. A rank result does not prove that a completing claim can be supplied without changing prices. These are boundaries of the conditional statements, not minor technical qualifications.

If a continuation-solvency boundary is locally $N = \underline{n}(z, x, K, \tau)$, diffusion must be tangent there and remaining drift must point inward. This local viability condition explains why a numerical box or conservative floor is not fiscal capacity. It also distinguishes a KKT residual, which values additional movement along an existing payoff, from an orthogonal residual, which requires a new payoff direction.

T3 Local Dynamics, Small Risk, and the Six-Cell Benchmark

Reader guide. This appendix records three nested analytical objects. The first is a deterministic worker-resource problem that defines fiscal wealth and the forward-looking tax trade-off. The second is a local quadratic approximation around a smooth deterministic interior anchor. The third is a small-risk comparison of public payoff menus. The first two organize local dynamics; the third separates marketed from unmarketed fiscal exposure. None supplies global existence, a fiscal-capacity frontier, or a calibrated policy recommendation.

T3.1 Deterministic separation and the tax costate

Let $\mathcal{F}(K, \tau)$ denote the worker-relevant flow of wages plus net rental-tax resources. Fiscal wealth is the value of choosing the tax-speed path:

$$J(K, \tau) = \sup_{\{\nu_t\}} \int_0^\infty e^{-\rho t} \left[\mathcal{F}(K_t, \tau_t) - \frac{Y(K_t)}{2\kappa_\tau} \nu_t^2 \right] dt, \quad \dot{K} = Kg(K, \tau), \quad \dot{\tau} = \nu. \quad (\text{T3.1})$$

The object is a worker-resource value: wages enter because the government values worker consumption, not because wages are Treasury collateral. On a selected smooth interior characteristic the HJB and tax-speed condition are

$$\rho J = \mathcal{F} + KgJ_K + \frac{\kappa_\tau}{2Y} J_\tau^2, \quad \nu = \frac{\kappa_\tau}{Y} J_\tau. \quad (\text{T3.2})$$

With $X = N + J$, the log benchmark gives $c^0 = \rho X$ and $V^0 = \rho^{-1} \log(\rho X)$. This is a deterministic separation, not an assertion that $X > 0$ is sufficient for borrowing or continuation feasibility.

Writing $m = J_\tau$, the costate equation is

$$\dot{m} = \rho m + \mathcal{B}(J_K - 1). \quad (\text{T3.3})$$

If the discounted terminal term vanishes, integration gives

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty e^{-\rho(u-t)} \mathcal{B}_u [1 - J_K(S_u)] du. \quad (\text{T3.4})$$

The direct fiscal benefit of a higher future rate is $\mathcal{B}_u$; the opposing term is the fiscal value of preserving capital. The identity is conditional on its characteristic, terminal condition, positive denominator, and slack constraints. It is not an impact tax rule or a universal sign result.

T3.2 Local quadratic dynamics and small risk

Around a deterministic interior anchor, collect productivity, automation, log capital, and the inherited tax-rate deviation in $y = (z, x, k, \tilde{\tau})'$. The local state system has the form

$$\dot{y} = Ay + B_\nu \nu, \quad \mathcal{L}_{\text{quad}}(y, \nu) = -\frac{1}{2} y' Q^L y - \frac{1}{2\chi_{\text{LQ}}} \nu^2, \quad (\text{T3.5})$$

after the primitive resource and capital equations have been differentiated at the anchor. The loss-side curvature H^L satisfies the discounted algebraic Riccati equation

$$\rho H^L = Q^L + A' H^L + H^L A - H^L B_\nu \chi_{\text{LQ}} B_\nu' H^L. \quad (\text{T3.6})$$

The economically relevant solution is the real symmetric stabilizing solution: it makes the closed-loop matrix $A_c = A - B_\nu \chi_{\text{LQ}} B_\nu' H^L$ Hurwitz. The qualifier matters. A Riccati root that does not stabilize the continuation does not describe the local Ramsey branch.

This module uses Q^L and H^L for the flow-loss and value-loss curvature objects. The extended theory derivation uses fiscal-value curvature $H^J = -H^L$ and flow-value curvature $Q^J = -Q^L$. With those sign maps and the common adjustment coefficient χ_{LQ} , the two Riccati representations are the same equation.

The capital-tax block is two dimensional. Partition the state as $y = (y_e', y_i')'$, where $y_e = (z, x)'$ contains the exogenous technology states and $y_i = (k, \tilde{\tau})'$ contains log capital and the inherited tax deviation. The block-triangular drift lets the curvature be computed in economic order. First solve the two-dimensional algebraic Riccati equation for H_{ii}^L and select the root whose closed-loop intrinsic drift is stable. Holding that block fixed, the cross curvature H_{ei}^L solves a Sylvester equation whose known forcing terms collect the primitive technology-to-capital drift and flow cross-

derivatives. Finally, H_{ee}^L solves a discounted Lyapunov equation using the already determined cross block. The fully expanded matrices appear in the extended original-coordinate derivation below. This ordering transmits persistent shocks into capital and tax adjustment and then values their continuation effects; it is not a global solution of the nonlinear Ramsey problem.

Strict additive-noise LQG is certainty equivalent: additive, control-independent variance shifts the value constant but does not change the Riccati feedback. Hence a precautionary tax-speed or portfolio correction cannot be inferred merely from a local LQG calculation. It belongs to the separate order-two small-risk expansion below.

The small-risk portfolio and welfare. Scale Brownian loadings and market prices of risk by ε . Let $\zeta_J = (\hat{\sigma}_z J_z, \hat{\sigma}_x J_x)'$ be fiscal-wealth exposure. For the aligned one-claim closure,

$$h_J = \frac{\zeta_J' \hat{\sigma}_R}{\|\hat{\sigma}_R\|^2}, \quad s^0 = X - h_J, \quad \zeta_J^\perp = \zeta_J - h_J \hat{\sigma}_R. \quad (\text{T3.7})$$

The first term in s^0 is compensated return demand; $-h_J$ offsets the fiscal exposure already delivered by the claim. The residual is orthogonal to the payoff. This leading position is selected by order- ε^2 return, variance, and covariance terms even though it is order one in dollars. A binding holding constraint requires a KKT/active-set treatment rather than clipping this formula after the fact.

The regular value expansion is

$$V^\varepsilon = V^0 + \varepsilon^2 V^2 + O(\varepsilon^4). \quad (\text{T3.8})$$

There is no first-order value term around the mean-zero deterministic reference. Conditional on common deterministic paths and feasible positions, the added value of a completing payoff is

$$V^{\text{complete}} - V^{\text{one}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\|\zeta_{J,t}^\perp\|^2}{2\rho X_t^2} dt + O(\varepsilon^4). \quad (\text{T3.9})$$

The exact projection is a geometric statement; this is a local welfare statement. It requires regularity, optimizer selection, transversality, an unchanged active set, and distance from economic constraints. It is not a finite-risk welfare magnitude.

T3.3 The six-cell benchmark

The six cells compare common inherited states and primitives. Tax regime $a \in \{F, A\}$ denotes frozen inherited taxation or Ramsey-adjustable taxation; payoff menu $m \in \{0, 1, 2\}$ denotes no risky claim, the maintained claim, or a locally completing two-claim menu. The cells are $(F0, F1, F2, A0, A1, A2)$. Within a tax row, the deterministic allocation is common across the three menus because risk vanishes at the deterministic limit. Across rows, fiscal wealth differs: frozen taxation solves a lower-dimensional characteristic with τ_0 held fixed, while adjustable taxa-

tion uses the tax-speed problem above.

Let P_m denote projection onto menu m . The row- a residual is $(I - P_m)\zeta_{J,a}$, and the order-two portfolio source is the squared marketed component scaled by worker risk tolerance. Thus the within-row one-claim and completion increments isolate successively the maintained and orthogonal payoff directions. Feasible-set nesting signs the exact total value difference only when the same inherited state is feasible in both cells. The order-two tax-asset interaction need not have a sign because the frozen and adjustable rows have different fiscal wealth, deterministic paths, and resolvents.

What the benchmark establishes. The deterministic separation and forward-tax identity are conditional deterministic characterizations. The Riccati/Sylvester/Lyapunov system is a local stabilizing approximation around its interior anchor. The projection identity is exact linear geometry for the declared local payoff map. The portfolio and welfare statements are regular small-risk results, not global or finite-risk results. The six-cell equations are conditional smooth-interior characterizations; their numerical magnitudes require separate accepted common-state computations. None of these modules proves a natural debt limit, household market completion, a first-best allocation, or an optimal calibrated policy.

T4 Full Brownian Ramsey Recursion and Conditional Solvency

Reader guide. This appendix defines the recursive Brownian Ramsey object used by the paper's conditional results. It distinguishes the economic closure of the five-state continuation problem from the unproved global existence and verification questions. The state is $S = (z, x, K, N, \tau)$; no promise component is generated under the maintained Brownian closures, but restart equivalence still requires a stopping-time-stable admissibility convention.

T4.1 Sequential economy and recursive state

Workers are hand to mouth and are the only agents with direct welfare weight. Domestic capitalists own installed capital and determine its accumulation through their after-tax return. International investors price the safe asset and one external risky claim. The government chooses worker consumption c , a signed dollar claim holding s , and tax speed $\nu = \dot{\tau}$. Installed capital and the tax rate are inherited at a Brownian innovation; the claim position is the direct public state-contingent instrument.

The date-zero objective is

$$V^R(S_0) = \sup_{(\Pi, E) \in \mathcal{R}(S_0)} \mathbb{E}_0 \int_0^\infty e^{-\rho t} \log c_t dt. \quad (\text{T4.1})$$

The admissible correspondence includes the private equilibrium, public budget, instrument bounds, production-regime conditions, integrability, and conditional continuation restrictions. The worker budget and government budget must remain separate: wages enter worker consumption and fiscal-resource value, but they are not Treasury cash receipts.

T4.2 Admissibility, solvency, and restart

An admissible policy is progressively measurable, makes every stochastic integral and adjustment cost integrable, respects the transfer, tax, and portfolio constraints, and satisfies no-Ponzi and terminal restrictions. Continuation solvency is stronger than a pointwise restriction such as $N \geq 0$: it asks whether at least one feasible tail can satisfy future resources, instruments, and conditional no-Ponzi restrictions. A solvency frontier, if it exists as a graph, is therefore schematic,

$$N \geq \underline{n}^{\text{sol}}(z, x, K, \tau). \quad (\text{T4.2})$$

It is distinct from a conservative numerical reference floor and from artificial box faces.

For every stopping time θ , restriction takes the tail of an admissible date-zero plan and views it as a continuation from S_θ . Pasting combines a feasible prefix with a measurable feasible continuation. The maintained five-state restart result is conditional on the same laws, prices, instruments, welfare objective, state-based solvency convention, conditional terminal conditions, and an admissible class stable under both operations. Under those hypotheses the date-zero optimal tail is optimal for the recursive continuation problem. Equality of values does not imply that the same controls are replayed: exact control replay also requires unique continuation optimization, equilibrium/branch selection, and closed-loop dynamics.

T4.3 Level HJB and term map

For a smooth candidate $V(z, x, K, N, \tau)$, write $\mathcal{L}^{z,x}$ for the drift and second derivatives of the two exogenous Brownian states. The smooth interior level HJB is

$$\begin{aligned} \rho V = \max_{c,s,\nu} \{ & \log c + \mathcal{L}^{z,x} V + KgV_K + \nu V_\tau \\ & + [rN + s\beta_R + \tau B + W - c - Y\nu^2/(2\kappa_\tau)]V_N \\ & + s[\sigma_z\sigma_{Rz}V_{Nz} + \sigma_x\sigma_{Rx}V_{Nx}] + \tfrac{1}{2}s^2\|\sigma_R\|^2V_{NN} \}. \end{aligned} \quad (\text{T4.3})$$

The first line propagates exogenous states, capital, and the inherited tax. The second is the public cash-flow drift. The last line is the covariance created because public net worth and the primitive states share Brownian shocks. It is an exact smooth-branch reduction, not a global verification theorem.

Pointwise interior recovery gives

$$c = V_N^{-1}, \quad \nu = \frac{\kappa_\tau}{Y} \frac{V_\tau}{V_N}, \quad s = -\frac{\beta_R V_N + D[V]}{\|\sigma_R\|^2 V_{NN}}. \quad (\text{T4.4})$$

These expressions require $V_N > 0$, $V_{NN} < 0$, and nonzero claim variance. At a portfolio bound, the final expression is replaced by the appropriate one-sided KKT condition. At the tax ceiling, tax speed lies in the tangent cone. At a transfer, solvency, or production boundary, the relevant object is a constrained or state-constraint HJB, not the unconstrained formula.

T4.4 Boundary and verification obligations

At a smooth solvency boundary, the diffusion of boundary distance must vanish and its remaining drift must point inward. These local viability conditions do not identify the maximal boundary. A completed classical verification theorem would need a well-defined viable domain, regularity of the candidate value, recovered controls that are admissible and keep the state in that domain, integrability of the value process, the required transversality condition, and a supermartingale/martingale argument for all admissible policies and the candidate policy. It would also need the production regime and every active constraint treated on its actual branch.

Those obligations are deliberately not asserted here. The maintained conditional result says what the five-state Bellman representation means if its solvency and admissibility hypotheses hold. It does not prove global feasible-set equality, existence of an optimum, uniqueness, an invariant distribution, or a natural debt limit.

References

- Acemoglu, Daron. 2025. “The Simple Macroeconomics of AI.” *Economic Policy* 40(121): 13–58.
- Acemoglu, Daron, and Pascual Restrepo. 2019. “Automation and New Tasks: How Technology Displaces and Reinstates Labor.” NBER Working Paper 25684.
- Aghion, Philippe, Benjamin F. Jones, and Charles I. Jones. 2019. “Artificial Intelligence and Economic Growth.” In *The Economics of Artificial Intelligence: An Agenda*, 237–282. University of Chicago Press.
- Andrews, Isaiah, and Maryam Farboodi. 2025. “Do Markets Believe in Transformative AI?” NBER Working Paper 34243, revised July 2026.
- Aparisi de Lannoy, Léo, Anmol Bhandari, David Evans, Mikhail Golosov, and Thomas J. Sargent. 2025. “Managing Public Portfolios.” *Journal of Political Economy* 133(12): 3903–3951.
- Bhandari, Anmol, David Evans, Mikhail Golosov, and Thomas J. Sargent. 2017. “Fiscal Policy and Debt Management with Incomplete Markets.” *Quarterly Journal of Economics* 132(2): 617–663.
- Farhi, Emmanuel. 2010. “Capital Taxation and Ownership When Markets Are Incomplete.” *Journal of Political Economy* 118(5): 908–948.
- Fleming, Wendell H., and H. Mete Soner. 2006. *Controlled Markov Processes and Viscosity Solutions*. 2nd ed. New York: Springer.
- Jones, Charles I. 2026. “AI and Our Economic Future.” *Journal of Economic Perspectives* 40(3): 3–22.
- Jones, Charles I., and Christopher Tonetti. 2026. “Past Automation and Future A.I.: How Weak Links Tame the Growth Explosion.” Working paper, version 0.5.
- Korinek, Anton, and Lee Lockwood. 2026. “Public Finance in the Age of AI: A Primer.” NBER Working Paper 34873.
- Korinek, Anton, and Donghyun Suh. 2024. “Scenarios for the Transition to AGI.” NBER Working Paper 32255.
- Merton, Robert C. 1971. “Optimum Consumption and Portfolio Rules in a Continuous-Time Model.” *Journal of Economic Theory* 3(4): 373–413.
- Moll, Benjamin, Łukasz Rachel, and Pascual Restrepo. 2022. “Uneven Growth: Automation’s Impact on Income and Wealth Inequality.” *Econometrica* 90(6): 2645–2683.
- Reinhorn, Leslie J. 2019. “On Optimal Redistributive Capital Taxation.” *Journal of Public Economic Theory* 21(3): 460–487.

- Restrepo, Pascual. 2025. “We Won’t Be Missed: Work and Growth in the AGI World.” NBER Working Paper 34423.
- Trammell, Philip, and Anton Korinek. 2026. “Economic Growth under Transformative AI.” *Annual Review of Economics* 18(1).